ProPer: Proactivity Driven Personalized Agents for Knowledge Gap Navigation

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amazon Research Day

Motivation: Epistemic Intelligence

- Effective human - AI collaboration extends beyond assistance to shaping the **problem formulation**.
- In practice, users operate under **partial epistemic states**, where critical aspects of the task remain unarticulated.
- These unobserved aspects (“*unknown unknowns*”) are valuable aspects to add to the user’s knowledge.
- Current agents primarily operate within expressed inputs, focusing on clarification rather than **epistemic expansion**.

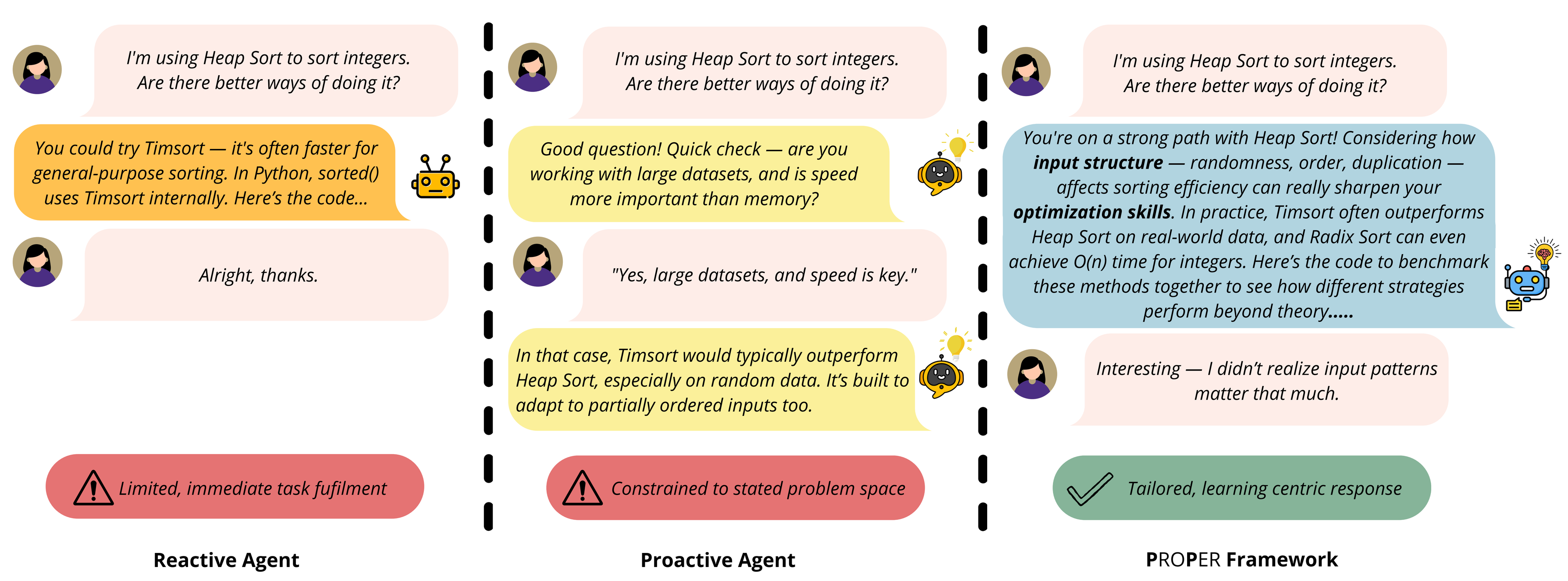
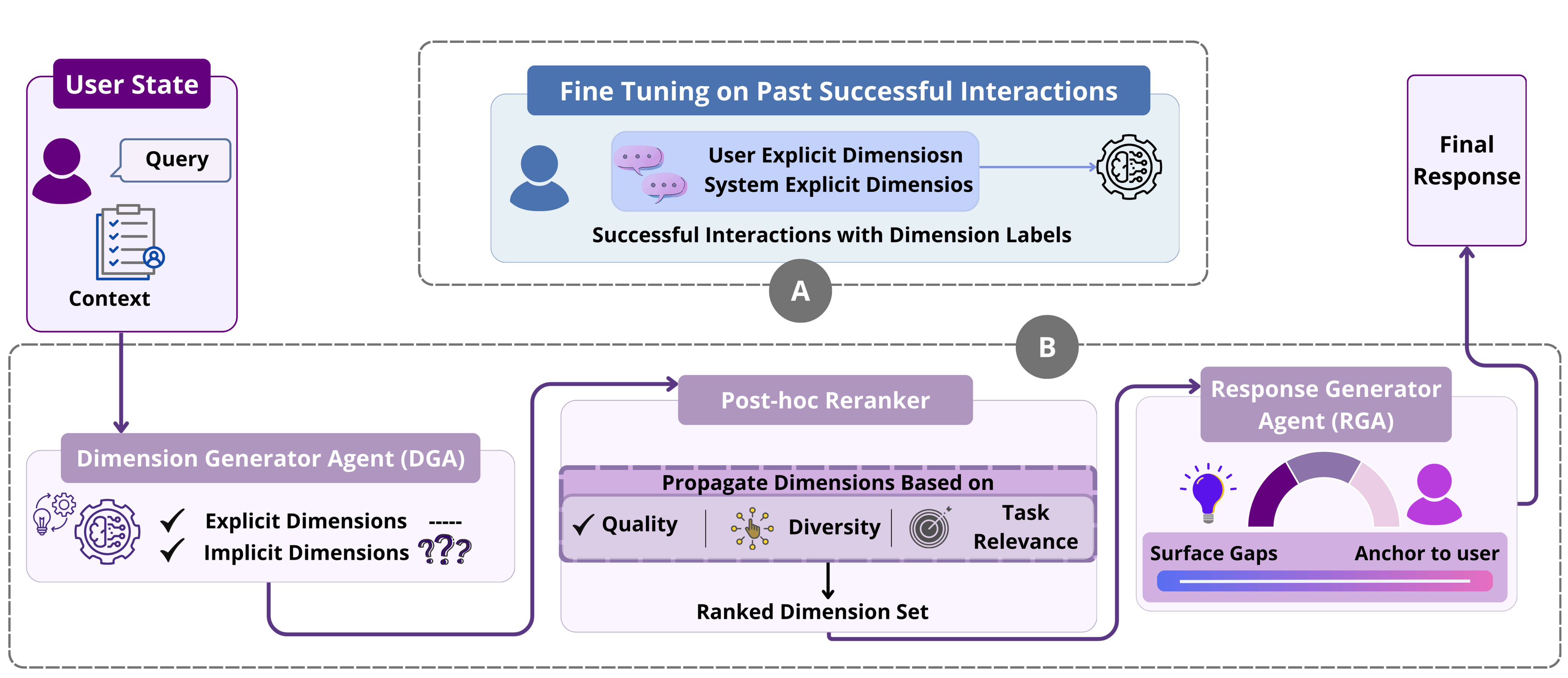


Figure 1. A comparison of various response paradigms approach to addressing a coding query.

Modeling the Missing

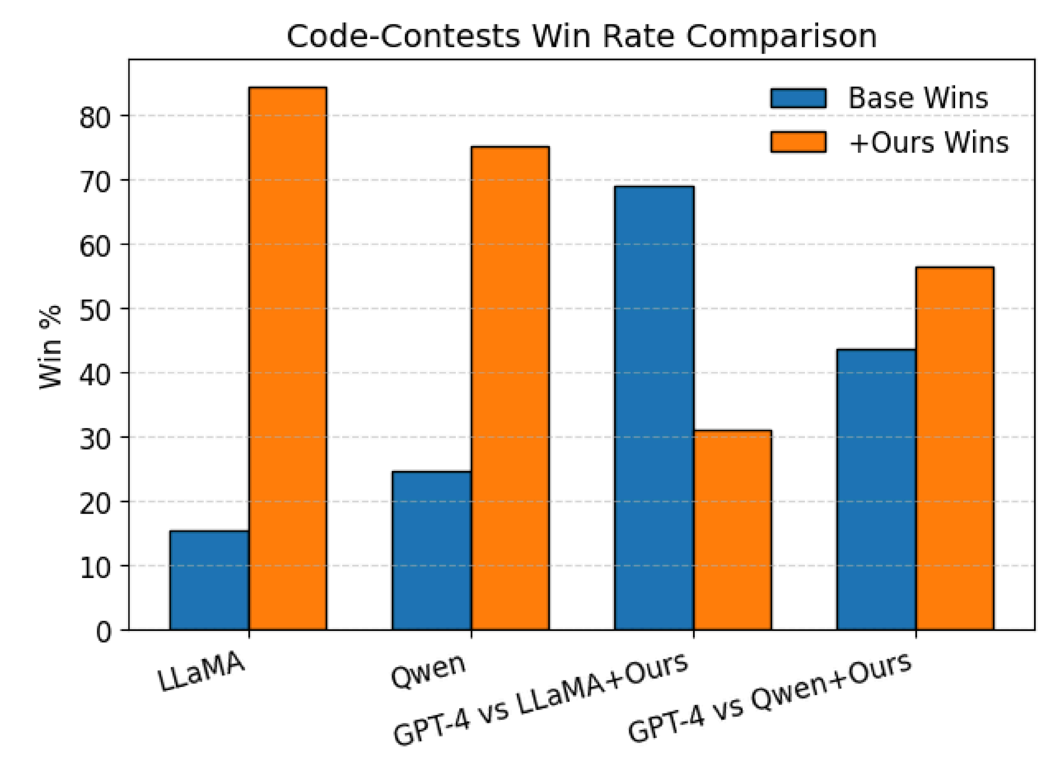
- Missing aspects can be viewed as **latent dimensions** (e.g., constraints, edge cases, alternatives)
- These dimensions can be inferred from expert models trained on domain knowledge.
- Not all inferred dimensions are useful.
- Effective collaboration requires selecting and contextualizing relevant dimensions.
- This enables personalized expansion of the problem space.

ProPer Framework



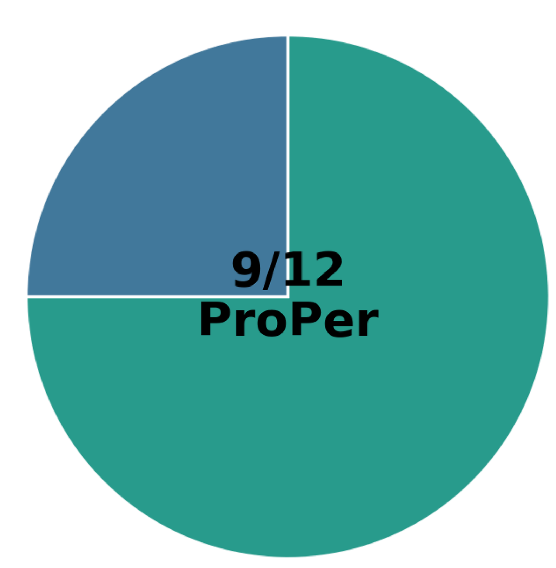
- Epistemic Modeling through dimension generation (DGA).
- Selective activation for relevance.
- Personalized response generation using the activated dimensions (RGA).

Empirical Evaluation on Coding Assistance



- LLM as Judge evaluation on Code Contest dataset
- +Ours wins 84.49% vs LLaMA and 75.24% vs Qwen → strong gains over base models.
- Against GPT-4: 56.37% (Qwen+Ours) and 31.07% (LLaMA+Ours) → competitive but mixed.
- Gains are smaller on Code-Contests, likely due to near-optimal solutions

Multi-Turn Evaluation



Preferred in 9/12 sampled multi-turn conversations, indicating consistent evaluator preference.

Key Takeaways

- Reframes proactivity as a calibration problem.
- Dimension-centric reasoning.
- Anticipation (what’s missing) matters more than generation capacity.
- Limited gains in tightly defined tasks.
- Shifts AI from reactive assistants → epistemically aware collaborators that help users think beyond what they asked.

Associated Papers

To appear at ACL’26

To appear at ICML’26

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